# 1A, 20V Schottky Barrier Surface Mount Rectifier

## FEATURES

- Low forward voltage, high efficiency
- High speed switching
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

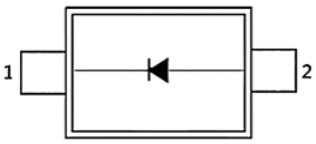
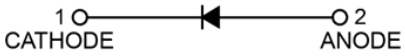
| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 1 | A |
| $V_{RRM}$ | 20 | V |
| $I_{FSM}$ | 5 | A |
| $T_{J\ MAX}$ | 125 | °C |

## APPLICATIONS

- Low stored charge
- For switching power supply
- Protection circuits

## MECHANICAL DATA

- Case: SOD-323F
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 1A whisker test
- Polarity: Indicated by cathode band
- Weight: 4.60mg (approximately)

| PACKAGE: SOD-323F | PIN CONFIGURATION | CIRCUIT DIAGRAM |
|---|---|---|
|  |  |  |



| ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted) | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **VALUE** | **UNIT** |
| Repetitive peak reverse voltage | $V_{RRM}$ | 20 | V |
| Reverse voltage, total rms value | $V_{R(RMS)}$ | 14 | V |
| Forward current | $I_F$ | 1 | A |
| Surge peak forward current 8.3ms single sine-wave superimposed on rated load | $I_{FSM}$ | 5 | A |
| Junction temperature | $T_J$ | -55 to +125 | °C |
| Storage temperature | $T_{STG}$ | -55 to +150 | °C |

| THERMAL PERFORMANCE | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **TYP** | **UNIT** |
| Junction-to-ambient thermal resistance | $R_{\Theta JA}$ | 220 | °C/W |

**Thermal Performance Note:** Units mounted on PCB (10mm x 5mm Cu pad test board)

## ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | CONDITIONS | SYMBOL | TYP | MAX | UNIT |
|---|---|---|---|---|---|
| Forward voltage(1) | $I_F$ = 10mA, $T_J$ = 25°C | $V_F$ | 266 | 290 | mV |
| | $I_F$ = 100mA, $T_J$ = 25°C | | 337 | 360 | mV |
| | $I_F$ = 1000mA, $T_J$ = 25°C | | 487 | 650 | mV |
| Reverse current @ rated $V_R$(2) | $V_R$ = 5V, $T_J$ = 25°C | $I_R$ | - | 10 | μA |
| | $V_R$ = 8V, $T_J$ = 25°C | | - | 20 | μA |
| | $V_R$ = 15V, $T_J$ = 25°C | | - | 50 | μA |
| Junction capacitance | f = 1MHz, $V_R$ = 5V | $C_J$ | 29 | 35 | pF |

**Notes:**

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

## ORDERING INFORMATION

| ORDERING CODE | PACKAGE | PACKING |
|---|---|---|
| BAT201M3 RRG | SOD-323F | 3,000 / 7” Tape & Reel |

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Forward Current Derating Curve**

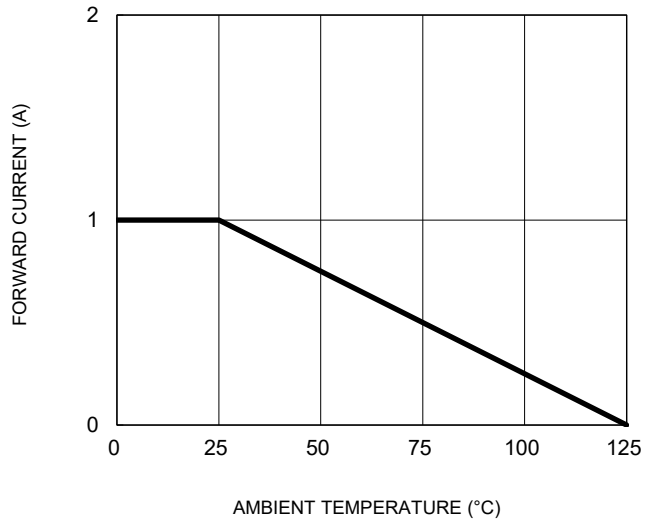



**Fig.2 Typical Junction Capacitance**

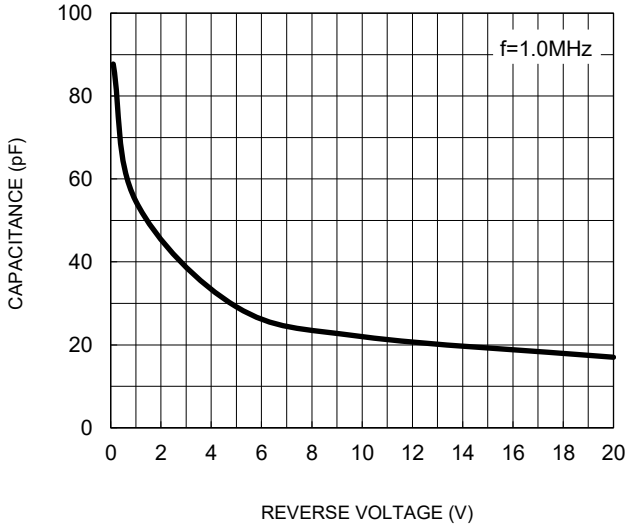



**Fig.3 Typical Reverse Characteristics**

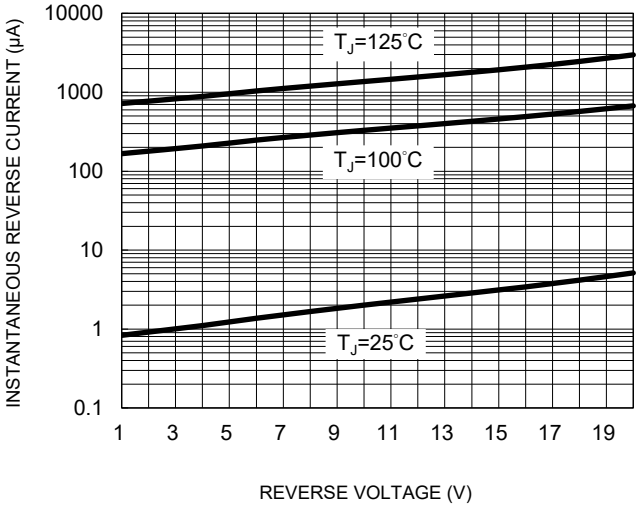
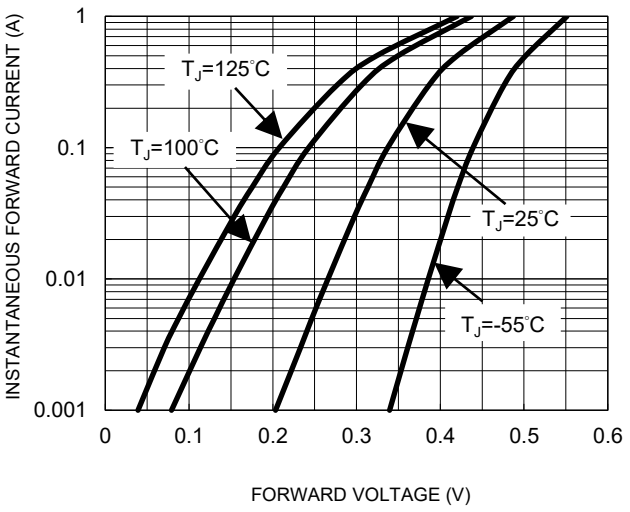
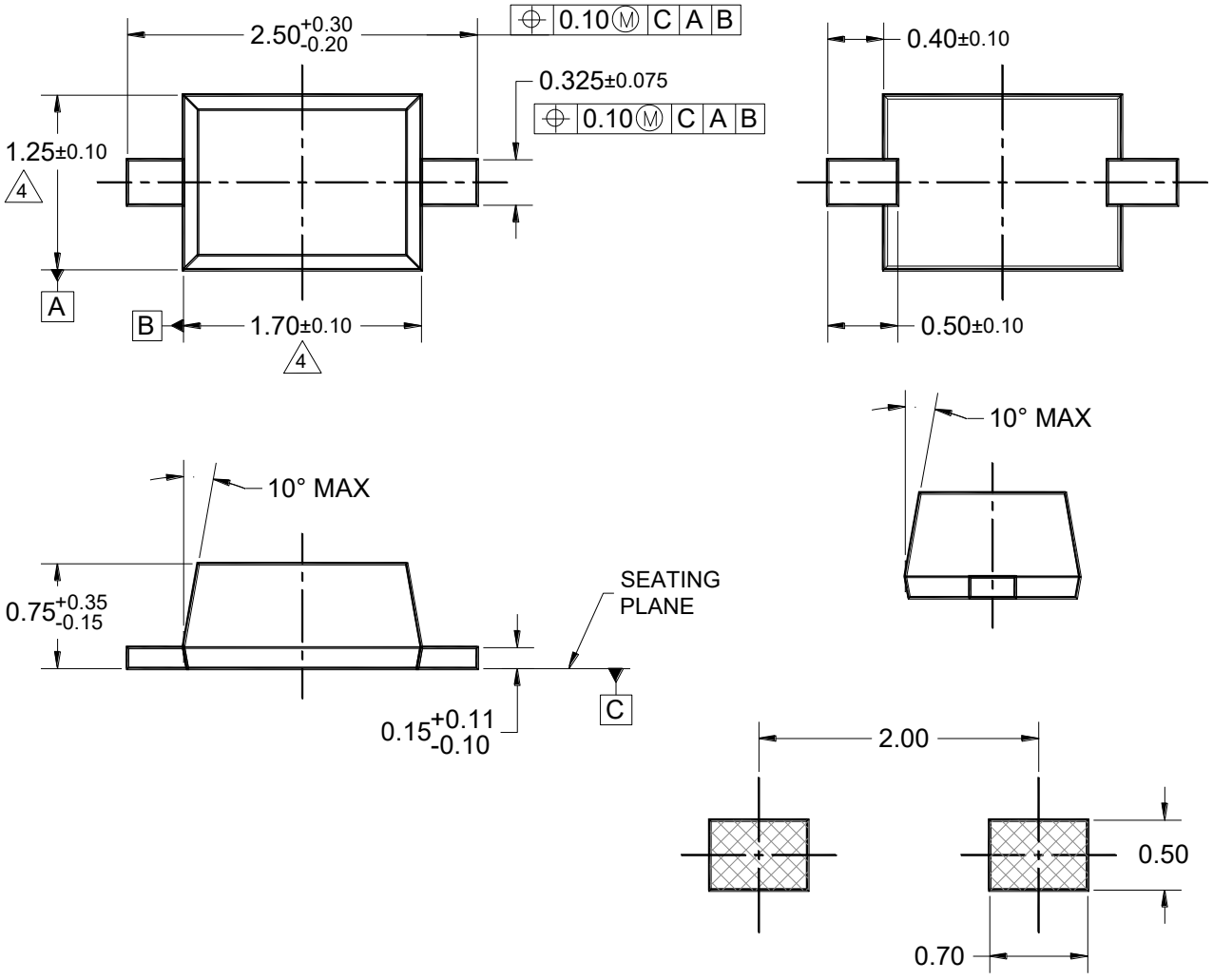



**Fig.4 Typical Forward Characteristics**

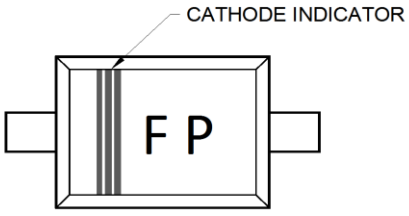


## PACKAGE OUTLINE DIMENSIONS

**SOD-323F**




SUGGESTED PAD LAYOUT




MARKING DIAGRAM

FP = Device marking

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: EIAJ ED-7500A-1, SC-90.
4. MOLDED PLASTIC BODY LATERAL DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-SOD323F-018 REV A.

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